

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 1.4 CONGRUENCE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use what you learned in all 8 lessons to answer the driving question: How does the mathematics of reflection explain why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy, and how can we use that same idea to prove that two shapes are truly identical? Write your answers in the space under each prompt. Use full sentences, show your working, and use these terms where they fit: line of reflection, perpendicular distance, image, congruence, and $y = mx + c$. Draw neat labelled diagrams where they help. Use the evidence you collected in class, such as your coordinate tables and the measurements you made, to support every claim.

Part 1: The Rule the Water Follows

At dawn near Kisumu, a fisherman's boat casts a mirror image on the still water of Lake Victoria. The image is flipped, but it looks exactly the same size as the boat. (a) Treat the waterline as the x-axis. Describe the rule that the water surface applies to every point of the boat, and write it as a coordinate rule. (b) Describe what you found when you measured the perpendicular distance from a vertex and from its image to the line of reflection. (c) Explain why this means the reflection is neither bigger nor smaller than the boat.

(a) The water surface acts as a line of reflection. When the waterline is the x-axis, every point of the boat is mapped by the rule $(x, y) \rightarrow (x, -y)$. A point 3 units above the waterline lands 3 units below it. (b) When we measured from each vertex of the boat polygon to the x-axis, and from each image vertex to the x-axis, the two perpendicular distances were always equal. The image point is on the opposite side of the line, directly across from the original. (c) Every point moves to the other side by exactly the same perpendicular distance, so no part of the boat is stretched or shrunk. The distances between points are kept, so the image has the same size and shape as the boat. It is only flipped, which is why the reflection is a perfect same-sized copy.

Part 2: Different Mirror Lines

The waterline will not always be on the x-axis. (a) Triangle ABC

(a) Across $y = 2$: the rule is $(x, y) \rightarrow (x, 2k - y)$ with $k = 2$, so $A(1, 3) \rightarrow (1, 4 - 3) = (1, 1)$. A is 1 unit above the line and the image is 1 unit below it. Across $x = 3$: the rule is $(x, y) \rightarrow (2k - x, y)$ with $k = 3$, so $A(1, 3) \rightarrow (6 - 1, 3) = (5, 3)$. A is 2 units left of the line and

has A(1, 3). Find the image of A when it is reflected across each of these lines: $y = 2$, $x = 3$, $y = x$, and $y = -x$. Show how you worked out each one. (b) Look at the four rules you used. What stays the same in all four reflections, even though the mirror line changes direction?	the image is 2 units right of it. Across $y = x$: the rule is $(x, y) \rightarrow (y, x)$, so $A(1, 3) \rightarrow (3, 1)$. Across $y = -x$: the rule is $(x, y) \rightarrow (-y, -x)$, so $A(1, 3) \rightarrow (-3, -1)$. (b) In every case the object point and its image are the same perpendicular distance from the line of reflection, on opposite sides. Because of this, the image is always congruent to the original figure. Diagonal reflections are not the same as rotations, even though they can look similar.
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Part 3: The Hidden Waterline	
One morning, mist hides the waterline. You can see a point on the boat, P(2, 5), and its mirror image P'(6, 5). (a) Explain how you can use only these two points to find the line of reflection. (b) Find the equation of the line of reflection. (c) Explain why your method works, using the idea of perpendicular distance.	(a) The line of reflection is the perpendicular bisector of the segment joining a point to its image. I join P to P', find the midpoint of that segment, and then draw the line through the midpoint at right angles to the segment. (b) The midpoint of PP' is $((2 + 6)/2, (5 + 5)/2) = (4, 5)$. The segment PP' is horizontal, so the perpendicular line is vertical. The equation of the line of reflection is $x = 4$. A vertical line has no gradient, so $x = 4$ is not of the form $y = mx + c$. The form $y = mx + c$ applies only to mirror lines that are not vertical. (c) The method works because every point and its image are the same perpendicular distance from the line of reflection. P is 2 units left of $x = 4$ and P' is 2 units right of it. The line therefore cuts the segment PP' exactly in half at a right angle, so it must be the perpendicular bisector. The same method can be used with the boat's other points to find a sloping mirror line, using the gradient and substitution to find $y = mx + c$.

Part 4: Proving Two Shapes Are Identical	
Triangle PQR on the boat is reflected in the waterline to give triangle P'Q'R'. (a) What does it mean mathematically for two shapes to be congruent? (b) Name the congruence condition you would use to prove that PQR is congruent to P'Q'R', and write out the steps of the proof. (c) Write the congruence statement with the vertices in the correct order. (d) List the other four congruence conditions.	(a) Two shapes are congruent if they are exactly the same in shape and size. One can be moved or flipped to fit exactly on top of the other. (b) I use SSS. Reflection keeps every point at an equal perpendicular distance from the waterline, so it keeps all distances. Step 1: $PQ = P'Q'$. Step 2: $QR = Q'R'$. Step 3: $PR = P'R'$. All three pairs of sides are equal, so the triangles are congruent by SSS. (c) Triangle PQR $\cong$ Triangle P'Q'R'. The vertices are written in matching order, so P matches P', Q matches Q', and R matches R'. (d) The other conditions are SAS (two sides and the angle between them), ASA (two angles and the side between them), AAS (two angles and a side not between them), and RHS (right angle, hypotenuse and one other side).

Part 5: Using the Idea in Real Life

Now answer the full driving question and apply it. (a) Write a clear explanation of why the mirror image of a boat on Lake Victoria is always a perfect, same-sized copy, and how the same idea proves two shapes are identical. (b) Give one real-life Kenyan example from the unit where reflection or congruence is used, and explain how the mathematics helps.	(a) The water surface acts as the line of reflection, and a boat's image is made by reflection. Every point of the boat is mapped to a point on the opposite side of the waterline at exactly the same perpendicular distance. Reflection is an isometry, which means it preserves all distances, so it never stretches, shrinks or distorts the boat. The waterline can be written as $y = mx + c$, or as the x-axis when it is at $y = 0$. Because all side lengths are kept, the image satisfies SSS and is congruent to the original boat: the same shape and size, only flipped. The same idea proves two shapes are identical. If we can show that matching sides and angles are equal using SSS, SAS, ASA, AAS or RHS, the shapes are congruent. (b) A surveyor in the Rift Valley can measure a gorge that cannot be crossed by making two congruent triangles. One triangle is on the surveyor's side of the gorge, and the other crosses it. Congruent triangles have equal matching sides, so a length that can be measured on land gives the width of the gorge. Another example is Maasai beadwork. Matching beads are the same distance from the central line of symmetry, so the design is balanced.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining reflection as the rule the water follows	Clearly states that every point and its image are the same perpendicular distance from the line of reflection, links this to the same size of the image, and gives correct coordinate rules.	States the equal-distance idea and gives the rule for the x -axis, but the link to the same size is not fully explained.	Says the image looks the same but gives no rule or no mention of equal perpendicular distance.
Accuracy of reflection calculations and the mirror line equation	All images are correct for every mirror line and the mirror line is found correctly as a perpendicular bisector, with working shown.	Most images are correct, and the mirror line is found with small errors in working.	Several images are wrong or the mirror line is not found.
Congruence proof	Names SSS, lists all three pairs of equal sides, writes the congruence statement in correct vertex order, and names the other four conditions.	Names the condition and writes the statement but leaves out a step or a condition.	Does not name a correct condition or does not give a congruence statement.
Use of evidence and mathematical vocabulary	Uses line of reflection, perpendicular distance, image, congruence and $y = mx + c$ correctly, and supports claims with values from class work.	Uses most of the terms correctly, with some claims given without evidence.	Uses few terms or uses them incorrectly, with little evidence.
Connecting to the phenomenon and real life	Gives a complete explanation of the Lake Victoria mirror image and a correct Kenyan application with clear reasoning.	Explains the mirror image and gives an application, but the reasoning is incomplete.	Gives a partial explanation, or an application with no reasoning.

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